Assignement 1

1. Finite Production

Hyundai's SUV division is selling only one product and is running an independent, fully automated facility with a production rate of 150 cars per month. The company needs 3 days of setup period to warm up the facility, and $1,000 is incurred daily for this task. It is facing a stable customer demand of 1,200 cars per year. The unit production cost is $15,000 and the selling price is $25,000. The company is applying an annual unit holding cost of 15% of product cost.

- Demand (D) = 1200 units / year
- Sell Price (p) = 25000 $ / unit
- Cost (c) = 15000 $ / unit
- Setup Cost (S) = 3000 $ / batch
- Production rate (P_{month}) = 150 units / month
- Production rate (P_{year}) = 1800 units / year
- Holding cost rate (h) = 15 % / unit / year
- Holding cost (H) = 2250 $ / unit / year

(a) *How many cars shall Hyundai produce whenever it starts the facility to maximize the total profit in the long run? (5 points)*

The total profit is given by the function:

$$\begin{aligned}\mathrm{TP}(Q) &= \mathrm{TR} - \mathrm{TC}(Q) \\ &= \underbrace{pD}_{\substack{\text{Total}\\\text{Revenue}}} - \underbrace{cD}_{\substack{\text{Purchasing}\\\text{Cost}}} - \underbrace{S\frac{D}{Q}}_{\substack{\text{Ordering}\\\text{Cost}}} - \underbrace{\frac{H}{2}\left(1-\frac{D}{P}\right)Q}_{\substack{\text{Holding}\\\text{Cost}}}\end{aligned}$$

Maximize the total profit w.r. to Q:

$$Q^* = \arg\max_Q \mathrm{TP}(Q)$$

Taking the derivative:

$$\begin{aligned}\frac{\mathrm{d}\ \mathrm{TP}}{\mathrm{d}Q} &= \frac{\mathrm{d}}{\mathrm{d}Q}\left[pD - cD - S\frac{D}{Q} - \frac{H}{2}\left(1-\frac{D}{P}\right)Q\right] \\ &= \frac{\mathrm{d}}{\mathrm{d}Q}[pD] - \frac{\mathrm{d}}{\mathrm{d}Q}[cD] - \frac{\mathrm{d}}{\mathrm{d}Q}\left[S\frac{D}{Q}\right] - \frac{H}{2}\left(1-\frac{D}{P}\right)\frac{\mathrm{d}}{\mathrm{d}Q}[Q] \\ &= 0 - 0 - \frac{\mathrm{d}}{\mathrm{d}Q}[SD\cdot Q^{-1}] - \frac{H}{2}\left(1-\frac{D}{P}\right)\cdot 1 \\ &= -SD\cdot\frac{\mathrm{d}}{\mathrm{d}Q}[Q^{-1}] - \frac{H}{2}\left(1-\frac{D}{P}\right) \\ &= -SD\cdot(-1)Q^{-2} - \frac{H}{2}\left(1-\frac{D}{P}\right) \\ &= S\frac{D}{Q^2} - \frac{H}{2}\left(1-\frac{D}{P}\right)\end{aligned}$$

Setting the derivative equal to zero to find the critical point:

$$\begin{aligned}S\frac{D}{Q^2} - \frac{H}{2}\left(1-\frac{D}{P}\right) &= 0 \\ \frac{H}{2}\left(1-\frac{D}{P}\right) &= \frac{DS}{Q^2} \\ Q^2 &= \frac{2DS}{H\left(1-\frac{D}{P}\right)} \\ Q &= \sqrt{\frac{2DS}{H\left(1-\frac{D}{P}\right)}}\end{aligned}$$

To confirm this critical point is a maximum rather than a minimum, check the second derivative:

$$\begin{aligned}\frac{\mathrm{d}^2\,\mathrm{TP}}{\mathrm{d}Q^2} &= \frac{\mathrm{d}}{\mathrm{d}Q}\left[\frac{DS}{Q^2} - \frac{H}{2}\left(1-\frac{D}{P}\right)\right] \\ &= \frac{\mathrm{d}}{\mathrm{d}Q}[DSQ^{-2}] - 0 \\ &= DS\cdot\frac{\mathrm{d}}{\mathrm{d}Q}[Q^{-2}] \\ &= DS\cdot(-2)Q^{-3} \\ &= -\frac{2DS}{Q^3}\end{aligned}$$

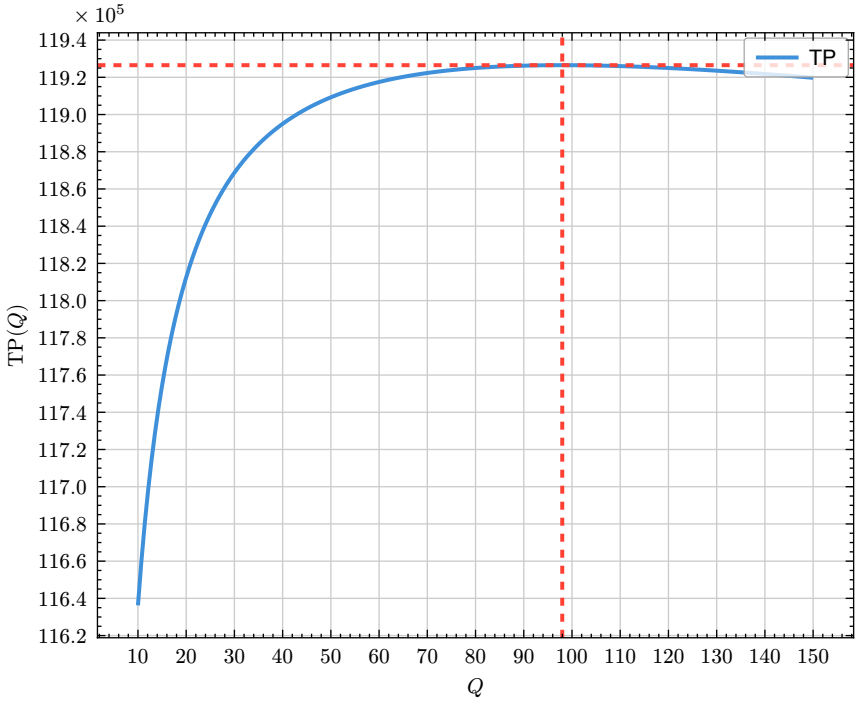
Since $D, S, Q > 0$, $\frac{\mathrm{d}^2\,\mathrm{TP}}{\mathrm{d}Q^2} < 0$ for all $Q > 0$, confirming $\mathrm{TP}(Q)$ is concave and the critical point $Q^* = \sqrt{\frac{2DS}{H(1-\frac{D}{P})}}$ is a maximum.

Using the formula we derived:

$$\begin{aligned}Q^* &= \sqrt{\frac{2DS}{H\left(1-\frac{D}{P}\right)}} \\ &= \sqrt{\frac{2(1200)(3000)}{2250 * \left(1-\frac{1200}{1800}\right)}} \\ &= \boxed{97.98}\end{aligned}$$

Therefore, the total profit:

$$\begin{aligned}\mathrm{TP}(Q^*) &= pD - cD - S\frac{D}{Q^*} - \frac{H}{2}\left(1-\frac{D}{P}\right)Q^* \\ &= (25000)(1200) - (15000)(1200) - 3000\frac{1200}{97.98} - \frac{2250}{2}\left(1-\frac{1200}{1800}\right)(97.98) \\ &= 11926515.31\end{aligned}$$



(b) *How many cars shall the company produce whenever it starts the facility to minimize the total cost in the long run? (5 points)*

The total cost is given by the function:

$$\mathrm{TC}(Q) = \underbrace{cD}_{\substack{\text{Purchasing}\\\text{Cost}}} + \underbrace{S\frac{D}{Q}}_{\substack{\text{Ordering}\\\text{Cost}}} + \underbrace{\frac{H}{2}\left(1-\frac{D}{P}\right)Q}_{\substack{\text{Holding}\\\text{Cost}}}$$

$$\mathrm{TVC}(Q) = \underbrace{S\frac{D}{Q}}_{\substack{\text{Ordering}\\\text{Cost}}} + \underbrace{\frac{H}{2}\left(1-\frac{D}{P}\right)Q}_{\substack{\text{Holding}\\\text{Cost}}}$$

Minimize the total cost w.r. to Q:

$$Q^* = \arg\min_Q \mathrm{TVC}(Q)$$

Find the first derivative of the total variable cost function:

$$\begin{aligned}
\frac{\mathrm{d}\ \mathrm{TVC}}{\mathrm{d}Q} &= \frac{\mathrm{d}}{\mathrm{d}Q}\left[S\frac{D}{Q} + \frac{H}{2}\left(1-\frac{D}{P}\right)Q\right] \\
&= \frac{\mathrm{d}}{\mathrm{d}Q}(DSQ^{-1}) + \frac{H}{2}\left(1-\frac{D}{P}\right)\cdot\frac{\mathrm{d}}{\mathrm{d}Q}(Q) \\
&= DS\cdot\frac{\mathrm{d}}{\mathrm{d}Q}(Q^{-1}) + \frac{H}{2}\left(1-\frac{D}{P}\right)\cdot 1 \\
&= DS\cdot(-1)Q^{-2} + \frac{H}{2}\left(1-\frac{D}{P}\right) \\
&= -\frac{DS}{Q^2} + \frac{H}{2}\left(1-\frac{D}{P}\right)
\end{aligned}$$

Setting the derivative equal to zero to find the critical point:

$$\begin{aligned}
-\frac{DS}{Q^2} + \frac{H}{2}\left(1-\frac{D}{P}\right) &= 0 \\
\frac{H}{2}\left(1-\frac{D}{P}\right) &= \frac{DS}{Q^2} \\
Q^2 &= \frac{2DS}{H\left(1-\frac{D}{P}\right)} \\
Q^* &= \sqrt{\frac{2DS}{H\left(1-\frac{D}{P}\right)}}
\end{aligned}$$

To confirm this critical point is a minimum rather than a maximum, check the second derivative:

$$\begin{aligned}
\frac{\mathrm{d}^2\ \mathrm{TVC}}{\mathrm{d}Q^2} &= \frac{\mathrm{d}}{\mathrm{d}Q}\left[-\frac{DS}{Q^2} + \frac{H}{2}\left(1-\frac{D}{P}\right)\right] \\
&= \frac{\mathrm{d}}{\mathrm{d}Q}[-DSQ^{-2}] + 0 \\
&= -DS\cdot\frac{\mathrm{d}}{\mathrm{d}Q}[Q^{-2}] \\
&= -DS\cdot(-2)Q^{-3} \\
&= \frac{2DS}{Q^3}
\end{aligned}$$

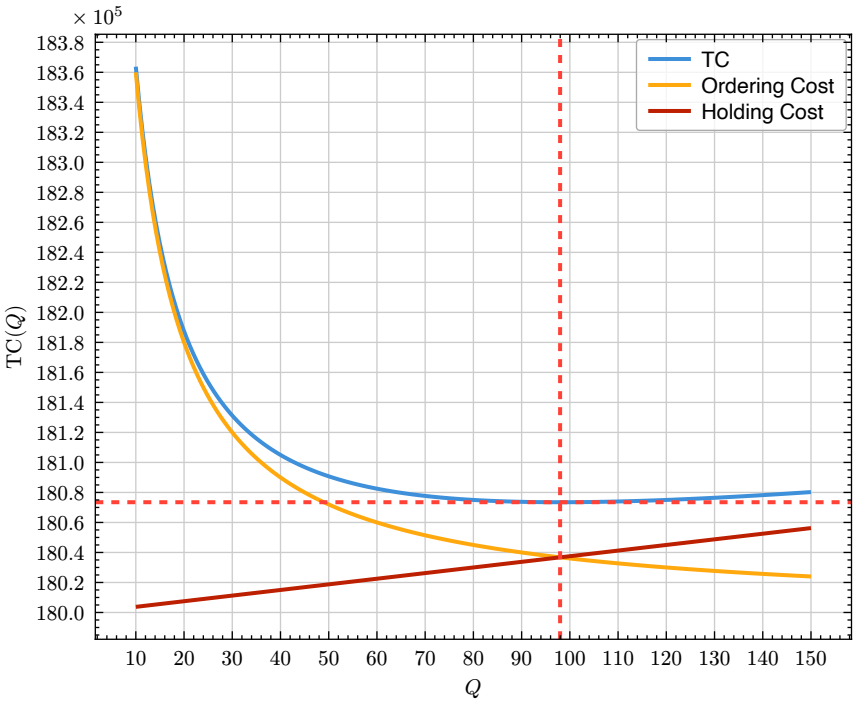
Since $D, S, Q > 0$, $\frac{\mathrm{d}^2\ \mathrm{TVC}}{\mathrm{d}Q^2} > 0$ for all $Q > 0$, confirming $\mathrm{TVC}(Q)$ is convex and the critical point $Q^* = \sqrt{\frac{2DS}{H\left(1-\frac{D}{P}\right)}}$ is a minimum.

Using the formula we derived:

$$\begin{aligned}
Q^* &= \sqrt{\frac{2DS}{H\left(1-\frac{D}{P}\right)}} \\
&= \sqrt{\frac{2(1200)(3000)}{2250 * \left(1-\frac{1200}{1800}\right)}} \\
&= \boxed{97.98}
\end{aligned}$$

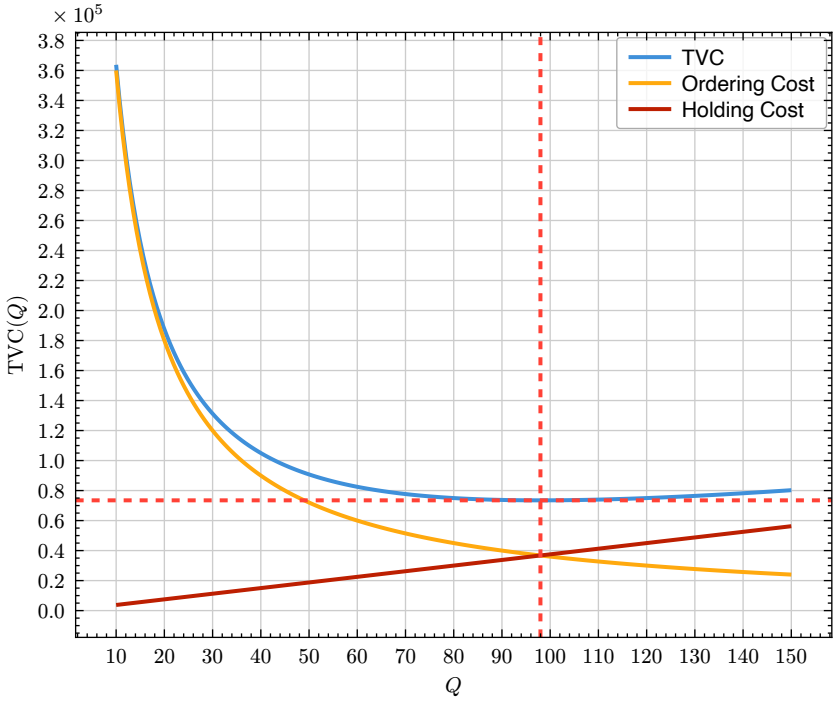
Therefore, the total cost:

$$\begin{aligned}\mathrm{TC}(Q^*) &= cD + S\frac{D}{Q^*} + \frac{H}{2}\left(1-\frac{D}{P}\right)Q^* \\ &= (15000)(1200) + 3000\frac{1200}{97.98} + \frac{2250}{2}\left(1-\frac{1200}{1800}\right)(97.98) \\ &= 18073484.69\end{aligned}$$



And total variable cost:

$$\begin{aligned}\mathrm{TVC}(Q^*) &= \sqrt{2DSH\left(1-\frac{D}{P}\right)} \\ &= \sqrt{2(1200)(3000)(2250)\left(1-\frac{1200}{1800}\right)} \\ &= 73484.69\end{aligned}$$



(c) *How many days per year shall the facility be idle for (b) above? (5 points)*

Assuming idle days do not include setup

$$T_c = T_s + T_p + T_i$$

Where:
- T_c: Cycle time
- T_s: Setup time
- T_p: Production time
- T_i: Idle time

$$T_p = \frac{Q}{P}$$

Number of production runs a year:

$$N = \frac{D}{Q^*} = \frac{1200}{97.98} = 12.25$$

Daily production:

$$p = \frac{1800}{365} = 4.93$$

Running (production) time per year

$$\frac{D}{p} = \frac{1200}{4.93} = 243.41$$

Each of the approximately 12.25 runs requires a 3 day setup:

$$N \times 3 = 12.25 \times 2 = 36.75$$

Idle time per year = Total days - running days - setup days

$$365 - 243.41 - 36.75 = \boxed{84.84}$$

Assuming idle days include setup

$$T_c = T_p + T_i$$

Where:
- T_c: Cycle time
- T_p: Production time
- T_i: Idle time

$$\begin{aligned} T_p &= \frac{Q}{P} \\ &= \frac{97.98}{1800} \\ &= 19.87 \end{aligned}$$

$$\begin{aligned} N &= \frac{D}{Q^*} \\ &= \frac{1200}{97.98} \\ &= 12.25 \end{aligned}$$

$$T_i = 365 - N \times T_p = \boxed{121.59}$$

Equivalently,

$$\begin{aligned} T_p &= \frac{Q}{P} \\ &= \frac{97.98}{1800} \\ &= 19.87 \end{aligned}$$

$$\begin{aligned} T &= \frac{Q^*}{D} \\ &= \frac{97.98}{1200} \\ &= 0.08165 \end{aligned}$$

$$\begin{aligned} T_d &= T - T_p \\ &= 0.08165 - 19.87 \\ &= 9.93 \end{aligned}$$

$$\begin{aligned} N &= \frac{D}{Q^*} \\ &= \frac{1200}{97.98} \\ &= 12.25 \end{aligned}$$

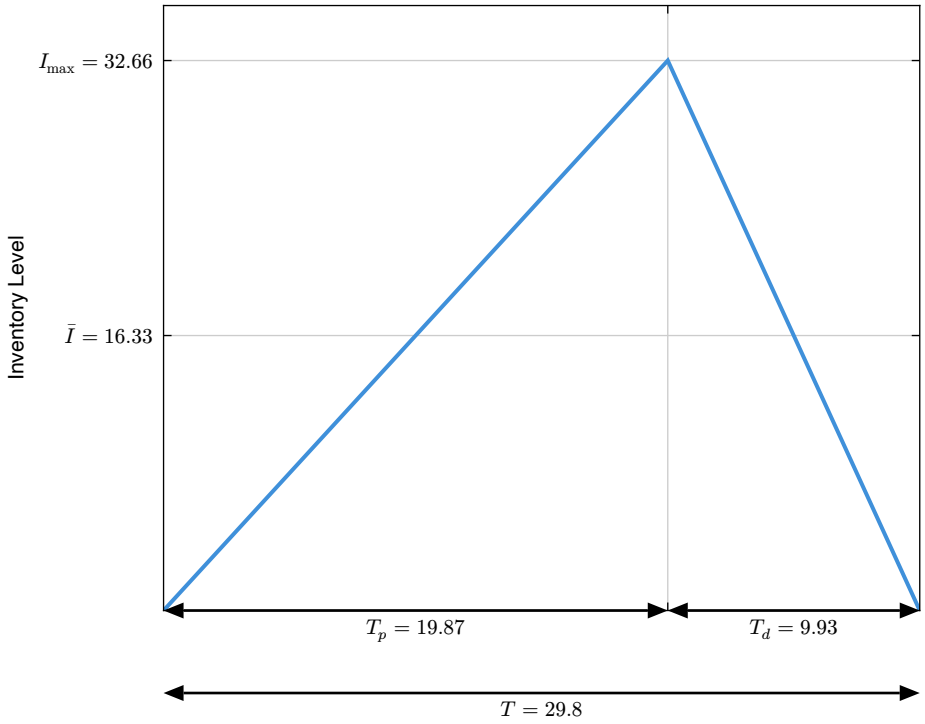
$$\boxed{121.64}$$

(d) *What is the average inventory level for (b) above in the long run? (5 points)*

Average inventory over the cycle is half of the peak:

$$\begin{aligned} I_{\max} &= Q^*\left(1 - \frac{D}{P}\right) \\ &= (97.98)\left(1 - \frac{1200}{1800}\right) \\ &= 32.66 \end{aligned}$$

$$\begin{aligned} \bar{I} &= \frac{I_{\max}}{2} \\ &= \frac{32.66}{2} \\ &= \boxed{16.33} \end{aligned}$$



(e) *How much is the average total profit per year in the long run? (5 points)*

$$\begin{aligned} \mathrm{TP}(Q^*) &= \mathrm{TR} - \mathrm{TC}(Q^*) \\ &= \underbrace{pD}_{\substack{\text{Total} \\ \text{Revenue}}} - \underbrace{cD}_{\substack{\text{Purchasing} \\ \text{Cost}}} - \underbrace{S\frac{D}{Q^*}}_{\substack{\text{Ordering} \\ \text{Cost}}} - \underbrace{\frac{H}{2}\left(1 - \frac{D}{P}\right)Q^*}_{\substack{\text{Holding} \\ \text{Cost}}} \\ &= (4.93)(1200) - (15000)(1200) - 3000\frac{1200}{97.98} - \frac{2250}{2}\left(1 - \frac{1200}{1800}\right)(97.98) \\ &= \boxed{11926515.31} \end{aligned}$$

(f) *Now the company has a new constraint on the production batch size and the production quantity should be a multiple of 40, i.e., 40 - 80 - 120 - 160 - …, whenever it runs the facility. How much is the optimal production quantity to minimize the long-run cost? (5 points)*

The constrained optimum must be one of the two feasible batch sizes bracketing Q^*: $Q = 80$ or $Q = 120$:

$$\mathrm{TC}(80) = cD + S\frac{D}{80} + \frac{H}{2}\left(1 - \frac{D}{P}\right)80 = \boxed{18075000}$$

$$\mathrm{TC}(120) = cD + S\frac{D}{120} + \frac{H}{2}\left(1 - \frac{D}{P}\right)120 = \boxed{18075000}$$

Both

$Q = \boxed{80}$

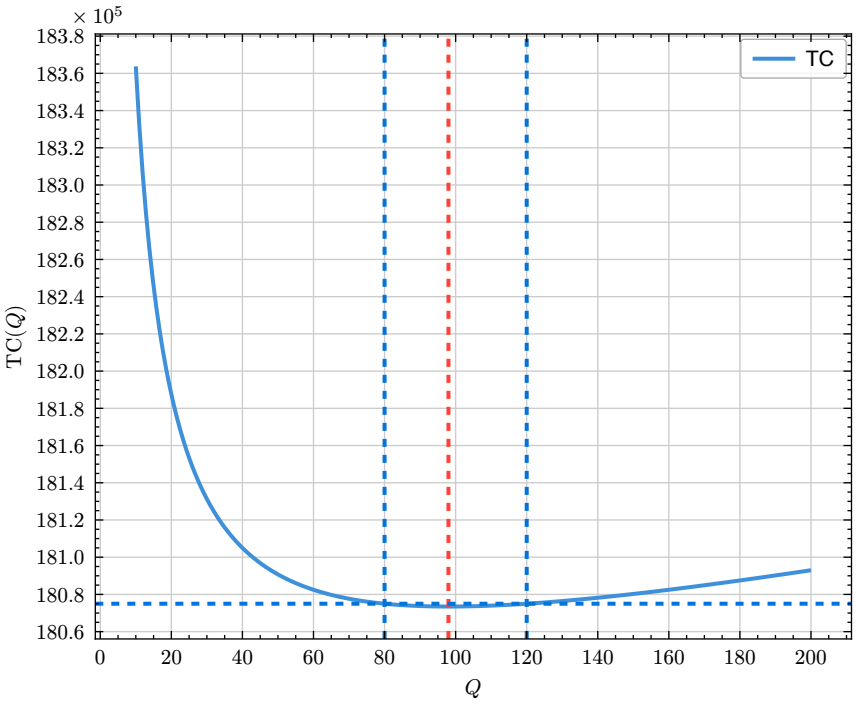
and

$Q = \boxed{120}$

are optimal, tied at:

$$\mathrm{TC}_{\min} = 18075000$$

which is 1515.31 $ higher per year than the unconstrained minimum of 18073484.69 $.



2. All unit and marginal unit discounts (Max Points 30)

Ericson's mobile communications division, a smartphone manufacturer, uses 20,000 memory chips per month. Their transportation service provider charges Ericson $300 per shipment, independently of the quantity purchased. The manufacturer offers an all unit quantity discount with a price of $1 per memory chip for orders up to 18,000 units, $0.96 per chip for orders between 18,000 and 36,000 units, and $0.92 per chip for orders larger than 36,000 units. Ericson incurs an annual unit holding cost of 15% of product cost.

- Demand D = 240000 units/year
- Setup cost S = $300/order
- Holding cost rate h = 15% of unit cost
- Break points: $q_1 = 18000$, $q_2 = 36000$
- Price c_i ($/unit): $c_1 = 1$, $c_2 = 0.96$, $c_3 = 0.92$

(a) *What is the optimal lot size for Ericson? (5 points)*

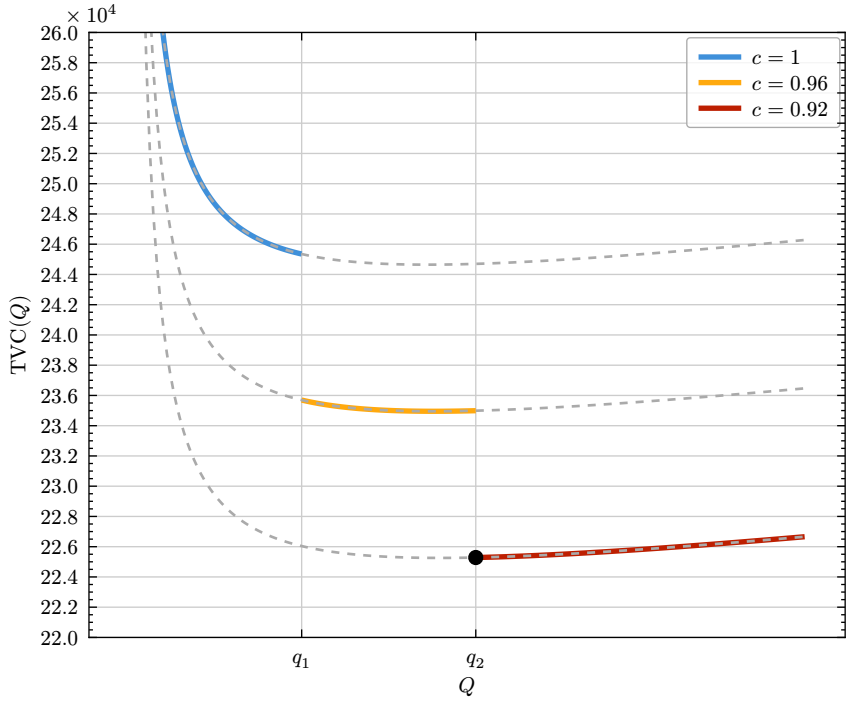
The total annual cost for bracket i is:

$$\mathrm{TVC}_i(Q) = Dc_i + \frac{D}{Q}S + \frac{Q}{2}hc_i$$

which is minimized at

$$\frac{\mathrm{dTVC}_i(Q)}{\mathrm{d}Q} = -\frac{DS}{Q^2} + \frac{hc_i}{2} = 0 \quad \Rightarrow \quad Q_i^* = \sqrt{\frac{2DS}{hc_i}}$$

Bracket	Range	Price	Unconstrained Q_i^*	Feasible?	Best Q in bracket	TVC_i
1	$0 \le Q \le 18000$	1	30983.87	No (too large)	18,000	245350
2	$18000 \le Q \le 36000$	0.96	31622.78	Yes	31622.78	234953.68
3	$Q \ge 36000$	0.92	32302.91	No (too small)	36,000	225284



$$Q^* = \boxed{36{,}000}$$

(b) *What is the annual cost of such a policy? (5 points)*

$$\mathrm{TVC}(Q^*) = \mathrm{TVC}(36000) = D(0.92) + \frac{D}{36000}(300) + \frac{36000}{2}(0.15)(0.92) = \boxed{225284}$$

(c) *What is the cycle inventory of memory chips at Ericson? (5 points)*

$$\frac{Q^*}{2} = \frac{36000}{2} = \boxed{18{,}000}$$

[(d)-(e)] *The manufacturer now offers a "marginal" unit quantity discount for the chips. The first 18,000 chips of any order are sold at $1 per unit, the next 18,000 units are sold at $0.96 and any quantity over 36,000 chips is sold for $0.92 per unit.*

(d) *What is the optimal lot size for Ericson given this pricing structure, what is the resulting cost, and how does it compare with that in (a) and (b)? (10 points)*

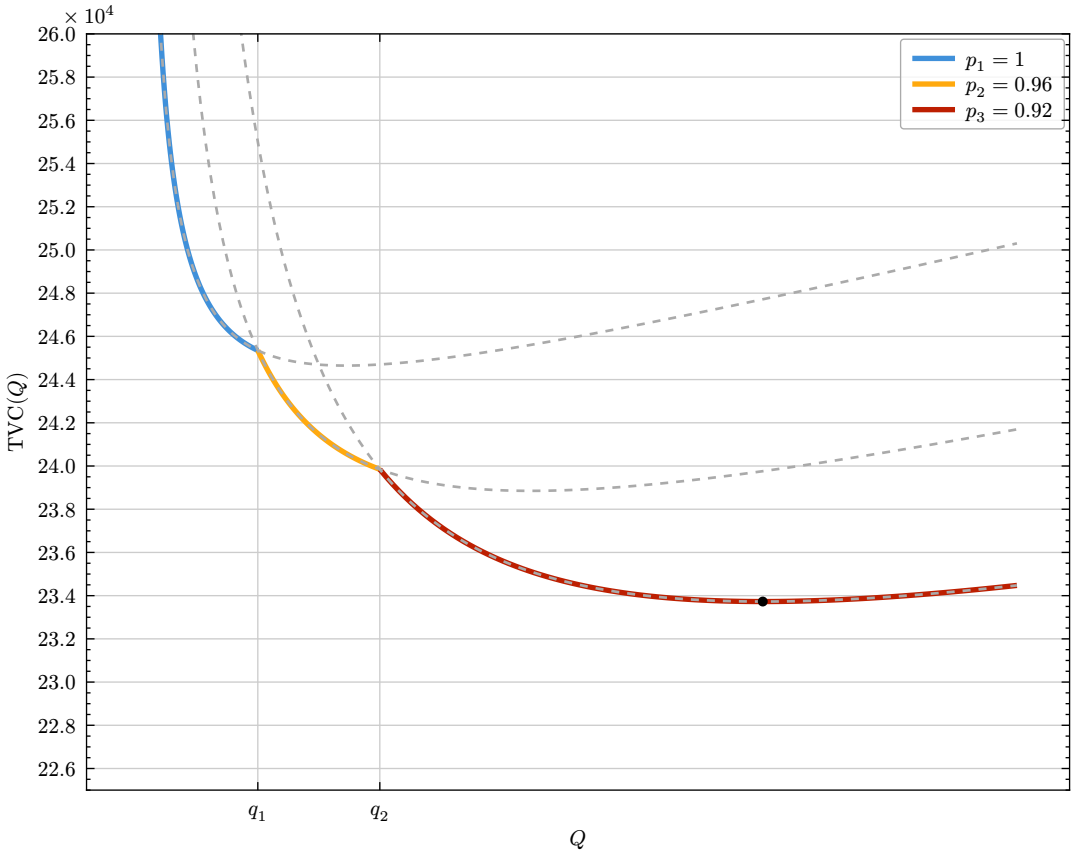
$$\begin{aligned} C_1(Q) &= Q && \text{for } \ 0 \le Q \le 18000 \\ C_2(Q) &= 18000 + 0.96(Q - 18000) && \text{for } \ 18000 \le Q \le 36000 \\ C_3(Q) &= 35280 + 0.92(Q - 36000) && \text{for } \ Q \ge 36000 \end{aligned}$$

$$\mathrm{TVC}_i(Q) = \frac{D}{Q}(S + C_i(Q)) + \frac{h}{2}C_i(Q)$$

$$\frac{\mathrm{dTVC}_i}{\mathrm{d}Q} = -\frac{D(S + R_i)}{Q^2} + \frac{hp_i}{2} = 0 \quad \Rightarrow \quad Q_i^* = \sqrt{\frac{2D(S + R_i)}{hp_i}}$$

Bracket	Range	Price p_i	R_i	Unconstrained Q_i^*	Feasible?	TVC_i
1	$0 \le Q \le 18000$	1	0	30983.87	No (too large)	245350

2	$18000 \leq Q \leq 36000$	0.96	720	58309.52	No (too large)	239846
3	$Q \geq 36000$	0.92	2160	92501.47	**Yes**	233727.2



$$Q^*_{\text{marginal}} = \boxed{92501.47}$$

(compare to the bracket boundaries 18000 and 36000)

$$\text{TVC}(Q^*_{\text{marginal}}) = \boxed{233727.2}$$

Comparison: the marginal-discount optimum requires ordering more than twice as much per shipment (92501.47 vs. 36,000) to reach a comparable cost level, and even so the resulting annual cost, $233727.2, is higher than the all-units discount's $225284.

(e) *What is the cycle inventory of memory chips at Ericson given the ordering policy? (5 points)*

$$\frac{Q^*_{\text{marginal}}}{2} = \frac{92501.47}{2} = \boxed{46250.74}$$

3. Shipment aggregation (Max Points 20)

LG, a refiner in Barcelona, serves three customers near Zaragoza, and maintains consignment inventory (owned by LG) at each location. Demand at the large customer is 48 tons a year, demand at the medium customer is 24 tons per year, and demand at the small customer is 12 tons per year. Product cost for LG is $80,000 per ton and the annual unit holding cost has been estimated to 15% of product cost. Currently, LG uses full-truckload transportation to deliver separately to each customer. Truck capacity is 12 tons and each delivery costs $700 plus $150 per stop (thus, delivering to each customer separately costs $850 per truck). LG is considering aggregating deliveries to Zaragoza on a single truck.

- Demand
  - Large (D_l) = 48 tons / year
  - Medium (D_m) = 24 tons / year
  - Small (D_s) = 12 tons / year
- Unit cost (C) = 80000 $ / ton
- Holding cost rate (h) = 15 %
- Holding cost ($H = Ch$) = 12000 $ / ton / year
- Truck capacity (M) = 12 tons / truck

- Major delivery cost (S) = 700 $ / delivery
- Minor stop cost (s_i) = 150 $ / stop
- Cost of a solo delivery ($S_i = S + s_i$) = 850 $ / truck

(a) *What is the annual transportation and holding cost if LG ships a full truckload each time a customer is running out of stock? How many days of inventory are carried at each customer under this policy? (5 points)*

Since every truck is full (12 tons):

- 4 trips/year to the large customer
- 2 trips/year to the medium customer
- 1 trips/year to the small customer

Customer	Deliveries/yr	Transport cost	Holding cost	Total cost
Large	4	$3400	$72000	$75400
Medium	2	$1700	$72000	$73700
Small	1	$850	$72000	$72850

Total annual cost = $221950

- Transportation $5950
- Holding $216000

Days of inventory carried (average inventory ÷ average daily demand, $= 365Q/(2D)$:

- large = 45.6 days
- medium = 91.3 days
- small = 182.5 days.

(b) *What is the optimal delivery policy to each customer if LG ships separately (not necessarily full truckloads) to each of them? What is the annual transportation and holding cost? How many days of inventory are carried at each customer under this policy? (5 points)*

Each customer order its own lot size

$$Q_i^* = \sqrt{\frac{2D_iS_i}{H}}$$

Customer	Q^* (tons)	Orders/yr	Days of inventory	Annual cost
Large	2.61	18.41	9.9	$31292
Medium	1.84	13.02	14	$22127
Small	1.3	9.2	19.8	$15646

Total annual (transportation + holding) cost = $69065

(c) *What is the optimal delivery policy to each customer if LG aggregates shipments to each of the three customers on every truck that goes to Zaragoza? What is the annual transportation and holding cost? How many days of inventory are carried at each customer under this policy? (10 points)*

Truck makes one route to all 3 customers per trip, so the fixed cost becomes:

$$S_{\text{agg}} = S + ns_i = 700 + (3)150 = 1150$$

over the combined demand $D_{\text{agg}} = 84$ tons/year

$$Q^*_{\text{agg}} = \sqrt{\frac{2D_{\text{agg}}S_{\text{agg}}}{H}} = 4.01 \text{ tons per trip}$$

Customer	Shipment size per trip (tons)	Days of inventory
Large	2.29	8.7

Medium	1.15	8.7
Small	0.57	8.7

They all carry the **same** number of days of inventory (8.7 days)

Total annual (transportation + holding) cost = $48150, versus $69065 under separate optimal delivery

A saving of $20915 (≈ 30.3%), and $173800 versus the original full-truckload policy.

4. Power of two policies (Max Points 20)

EasyWash is a washing machine manufacturer operating in Europe. One of its best-selling models is produced in the firm's manufacturing site in Zaragoza and consists of three core subassemblies: rotor, cylinder, and frame. While the rotor is manufactured in-house, the other two subassemblies (cylinder and frame) are sourced from Asia. The cost for placing an order with the cylinder supplier is 2,500$, while the same cost for the frame is 600$. Also, the annual cost of holding one unit of inventory is equal to 75$ for the cylinder and 50$ for the frame. The annual demand for the particular model is equal to 1,200 units (each machine consists of one rotor, one cylinder, and one frame).

Assuming that availability of rotor is not an issue, the firm would like to coordinate the procurement of the other two subassemblies (cylinder and frame) for logistics planning purposes, by using "power of two" policies. What would be the optimal policies for the two subassemblies, considering that the firm's planning team uses one week as the base planning period (TB)? For each subassembly separately, please evaluate your policy's performance (cost-wise) compared to the optimal one.

Base period (T_B) = 1 week

$$
\begin{aligned}
2^0 \cdot T_B &= 1 \text{ week} \\
2^1 \cdot T_B &= 2 \text{ week} \\
2^2 \cdot T_B &= 4 \text{ week} \\
2^3 \cdot T_B &= 8 \text{ week} \\
&\vdots
\end{aligned}
$$

Since:

$$Q = DT$$

Substitute $Q = DT$ into $\mathrm{TVC}(Q)$:

$$\mathrm{TVC}(Q) = \frac{SD}{Q} + \frac{HQ}{2}$$

Ordering term becomes:

$$\frac{SD}{Q} = \frac{SD}{DT} = \frac{S}{T}$$

Holding term becomes:

$$\frac{HQ}{2} = \frac{H(DT)}{2} = \frac{HD}{2}T$$

Therefore:

$$\mathrm{TVC}(T) = \frac{S}{T} + \frac{HD}{2}T$$

Taking the derivative:

$$\frac{\mathrm{d}\ \mathrm{TVC}(T)}{\mathrm{d}T} = -\frac{S}{T^2} + \frac{HD}{2} = 0 \quad \Rightarrow \quad T^* = \sqrt{\frac{2S}{HD}}$$

Our optimal cycle time (T^*) for *cylinders* is:

$$
\begin{aligned}
T_c^* &= \sqrt{\frac{2(2500)}{(75)1200}} \\
&= 0.24
\end{aligned}
$$

Convert from years to weeks:

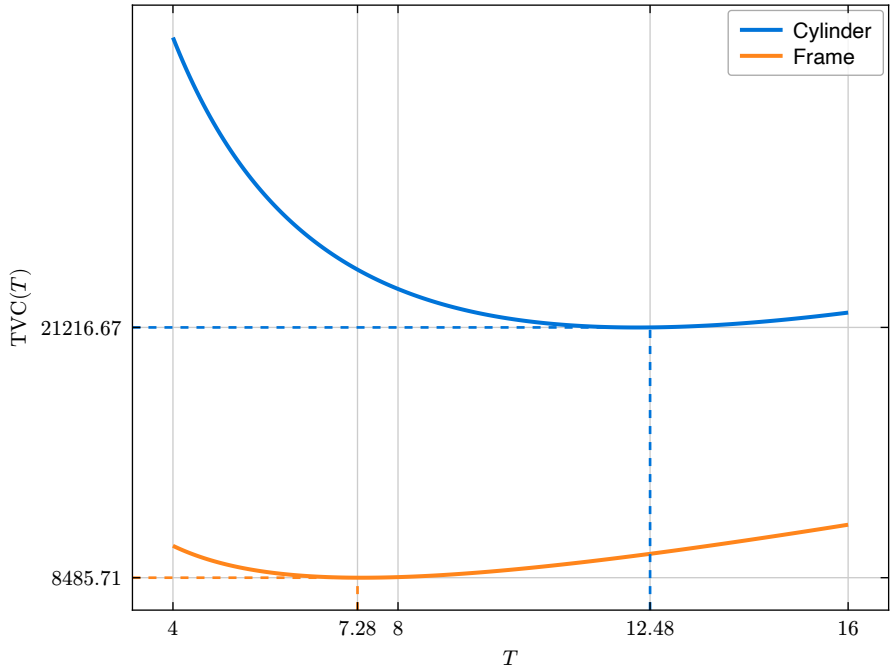
$$
\begin{aligned}
T_c^* \times 52 &= 0.24 \times 52 \\
&= 12.48
\end{aligned}
$$

Similarly for *frames*:

$$T_f^* = \sqrt{\frac{2(600)}{(50)1200}}$$
$$= 0.14$$

From years to weeks:

$$T_f^* \times 52 = 0.14 \times 52$$
$$= 7.28$$



We now want to find the power of 2 that braket T^*:

$$2^k T_B \leq T^* \leq 2^{k+1} T_B$$

For *cylinder*:

$$2^3 T_B \leq T_c^* \leq 2^4 T_B$$

Since T_c^* is closer to 16, we order every 16 weeks.

For *frame*:

$$2^2 T_B \leq T_f^* \leq 2^3 T_B$$

Since T_f^* is closer to 8, we order every 8 weeks.

Thus,

$$\boxed{T_c = 16 \text{ weeks}}$$

$$\boxed{T_f = 8 \text{ weeks}}$$

To compare the optimal with our new constrained schedule:

$$\text{TVC}(T_c^*) = 21216.67$$
$$\text{TVC}(T_c) = 21971.153846153848$$
$$\Delta = 0.036 < 0.06$$

And

$$\text{TVC}\left(T_f^*\right) = 8485.71$$
$$\text{TVC}\left(T_f\right) = 8515.384615384615$$
$$\Delta = 0.003 < 0.06$$

Quiz Problems

5. Batches Production (Max Points 10)

In the Economic Production Quantity (EPQ) model, we have assumed that output immediately becomes available to meet demand. Therefore, we can wait until inventory drops to zero before beginning production. In many cases, however, the finished product cannot be used until an entire batch is complete, perhaps because the batch must be moved to a warehouse, or because some finishing operation (e.g., cooling) is required. Let D = demand rate, P = production rate $(P > D)$, S = setup cost, H = unit holding cost, and Q = production quantity per production run.

(a) *Show that the average inventory, optimal production quantity, and total variable cost are respectively equal to:*

$$\bar{I} = \frac{1}{2}\left(1 + \frac{D}{P}\right), \quad Q^* = \sqrt{\frac{2DS}{H\left(1+\frac{D}{P}\right)}}, \quad \mathrm{TCV}(Q^*) = \sqrt{2DSH\left(1+\frac{D}{P}\right)}$$

(Hint: consider, to avoid running out of stock, what are the maximum and minimum inventory values for this system). (7 points)

A production run makes Q units at rate P, so it takes $t_1 = \frac{Q}{P}$ time to finish one run. Once it's finished, the whole batch Q drops into usable inventory at once. Demand keeps draining stock at rate D the whole time, in both phases. Since exactly Q units are consumed per cycle at rate D, a full cycle takes $T = \frac{Q}{D}$

1. Finished-goods stock behaves like ordinary EOQ

Usable (finished) inventory jumps up by Q the instant a batch finishes, then drains at rate D until the next batch finishes, $T = \frac{Q}{D}$ later. That's just a plain sawtooth from Q down to 0 (production rate P doesn't distort its shape, because nothing is added gradually). So:

$$\text{average finished-goods stock} = \frac{Q}{2}$$

2. But the batch being made also sits around and ties up cost

While a run is in progress ($t_1 = \frac{Q}{P}$ out of every cycle T), there's a pile of work-in-process building up in the factory. It starts at 0, grows linearly to Q by the end of the run, then disappears (converts to finished goods) for the rest of the cycle. That's a triangle of height Q over a width t_1, sitting inside a cycle of length T. Its time-average over the whole cycle is:

$$\frac{\frac{1}{2}\cdot Q \cdot t_1}{T} = \frac{Q}{2}\cdot\frac{t_1}{T} = \frac{Q}{2}\cdot\frac{Q/P}{Q/D} = \frac{Q}{2}\cdot\frac{D}{P}$$

This work-in-process still costs money to hold even though it can't be sold yet. That's what delayed-availability costs.

3. Add them up

$$\bar{I} = \frac{Q}{2} + \frac{Q}{2}\cdot\frac{D}{P} = \boxed{\tfrac{Q}{2}\left(1+\tfrac{D}{P}\right)}$$

4. Optimal Q

Total variable cost per unit time is holding cost plus setup cost:

$$\mathrm{TVC}(Q) = H\bar{I} + S\frac{D}{Q} = \frac{H}{2}\left(1+\frac{D}{P}\right)Q + \frac{SD}{Q}$$

Take the derivative and set it to zero:

$$\frac{\mathrm{d}\ \mathrm{TVC}(Q)}{\mathrm{d}Q} = \frac{H}{2}\left(1+\frac{D}{P}\right) - \frac{SD}{Q^2} = 0 \quad \Rightarrow \quad \boxed{Q^* = \sqrt{\tfrac{2DS}{H\left(1+\frac{D}{P}\right)}}}$$

5. Optimal cost

This is the standard $\mathrm{TVC}(Q) = aQ + \frac{b}{Q}$ shape, whose minimum value is $2\sqrt{ab}$. Here $a = \frac{H}{2}\left(1+\frac{D}{P}\right)$ and $b = SD$, so:

$$\mathrm{TVC}(Q^*) = 2\sqrt{ab} = \boxed{\sqrt{2DSH\left(1+\tfrac{D}{P}\right)}}$$

(b) *Notice that when output is immediate available, the total variable cost is increasing in the production rate, while in this case (i.e., when the finished product cannot be used until an entire batch is complete), the total variable cost is decreasing in the production rate. Explain the possible reason. (3 points)*

- **Immediate-use model** (cost rises with P)

while a batch is being made, units are sold off as fast as they're produced, so the pile never grows to the full size Q — it only reaches $Q\left(1-\frac{D}{P}\right)$, because selling at rate D "thins it out" the whole time it's being built. A slow production rate gives demand more time to eat into the pile during the run, keeping average inventory low. As P increases, the run gets shorter and there's less time for that thinning effect, so the peak — and the cost — creeps up toward the plain-EOQ value.

- **Delayed-availability model** (cost falls with P)

here nothing gets sold off while it's being made, so demand during the run has to be covered by stock (or WIP) held in reserve for exactly that window, of length $t_1 = \frac{Q}{P}$. A slow production rate means a long blackout window, which means a bigger reserve has to sit around the whole time — extra cost. As P increases, the run gets shorter, the blackout shrinks, and less reserve is needed, so cost falls toward the plain-EOQ value.

Conclusion

A finite production rate helps you when output is usable immediately (you get to sell while you build), but hurts you when output is stuck in limbo until the batch is done (you have to cover demand out of reserves for the whole build time) — which is exactly the sign flip between $\left(1 - \frac{D}{P}\right)$ and $\left(1 + \frac{D}{P}\right)$.

6. Quiz Problem 2. EOQ with Backorders (Max Points 10)

The Atlantic Coast Tire Corporation Problem

The Atlantic Coast Tire Corporation (ACT) is the east coast distributor of Eversafe tires supplying 1500 retail stores and service stations. Currently, ACT follows a simple policy for replenishing its inventory. When the inventory level of a particular size tire gets low, ACT places a large order with Eversafe to replenish its stock. Shipments arrive by truck 9 working days after the placement of the order.

The following are data for tire R13:
- These tires sell at a regular rate of about 500 per month; that is, 6,000 items per year.
- The current policy has been to order 1,000 tires as needed every couple months.
- The order is placed just in time to have the delivery arrive as inventory runs out.

Inventory Cost Components for R13:
- The purchasing cost for ATC is $20 per tire.
- The cost for placing an order is $115, which includes:
  - Cost per shipment of tires (consider this as independent of the number of tires in each shipment;
  - Administrative cost for initiating and processing the purchase order.
- The annual cost of holding inventory is $4.20 per tire, which includes:
  - The cost of capital tied up in inventory (estimated at 15% of unit cost per annum);
  - The cost of leasing warehouse space;
  - The cost of insurance against loss by fire, theft, etc.;
  - The cost of inventory personnel;
  - Taxes that are based on the value of inventory.
- The cost of being out of stock is $7.50 per tire short per year, based on these consequences:
  - Customer dissatisfaction;
  - Potential price drop to placate a customer due to late deliveries;
  - Delayed revenue;
  - The cost of additional record keeping required for out-of-stock tires.

- Demand (D): 6000 units / year
- Lead Time (L): 9 days
- Cost (c): 20 $ / unit
- Setup Cost (S): 115 $ / order
- Holding Cost (H): 4.2 $ / unit / year
- Stockout (Penalty) Cost (P): 7.5 $ / unit

(a) What is the total annual cost of the current policy for ACT?

$$\begin{aligned}\mathrm{TC}(Q_0) &= cD + S\frac{D}{Q_0} + H\frac{Q_0}{2} \\ &= 122790\end{aligned}$$

(b) What is the annual variable cost of the current policy for ACT?

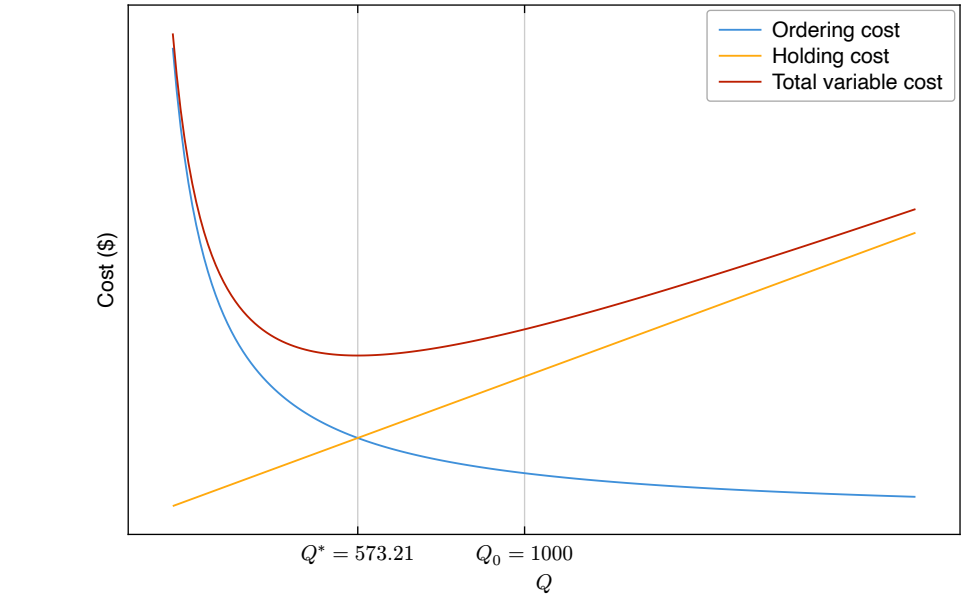
$$\begin{aligned}\mathrm{TVC}(Q_0) &= S\frac{D}{Q_0} + H\frac{Q_0}{2} \\ &= 2790\end{aligned}$$

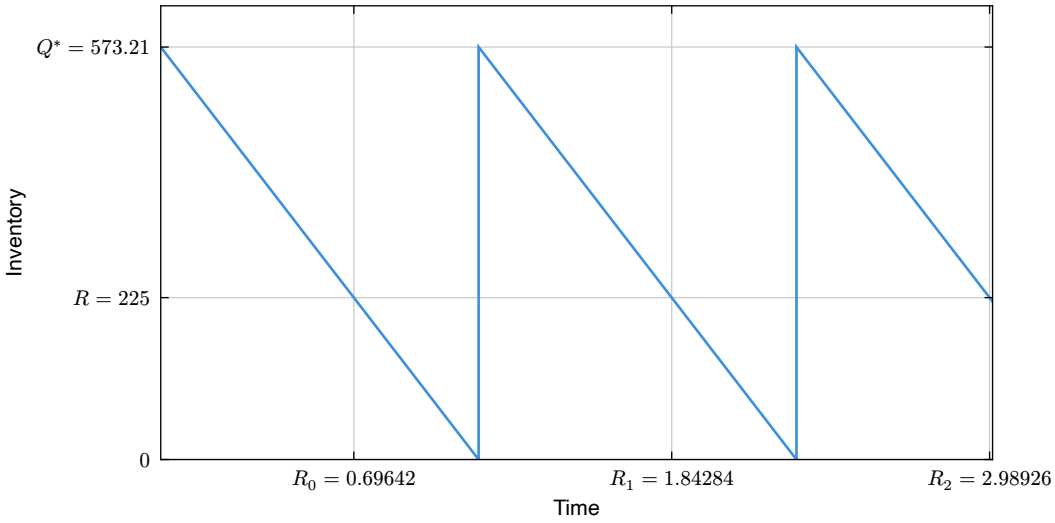
(c) Is the policy followed optimal? If not, how we could derive the optimal policy?

$$Q^* = \sqrt{\frac{2DS}{H}}$$
$$= 573.21$$

$$\mathrm{TC}(Q^*) = cD + S\frac{D}{Q^*} + H\frac{Q^*}{2}$$
$$= 122407.49$$

$$\mathrm{TVC}(Q^*) = S\frac{D}{Q^*} + H\frac{Q^*}{2}$$
$$= 2407.49$$





In the Atlantic Coast Tire Corporation (ACT) problem that we discussed during our first lecture on EOQ, we ignored the cost of being out of stock ($7.50 per tire short). Consider the penalty associated with stocking out as cost per unit short per year (i.e., the stockout cost penalizes also the duration of the stockout situation; hence it shall be treated as the holding cost). Assuming that all demand that is not satisfied on time is backordered (i.e., it is served, usually first, during the next batch delivery):

2 decision variables:
- Q: order quantity (same as before)
- b: the maximum backorder level ($0 \leq b \leq Q$)

In the classical EOQ:

$$\mathrm{TVC}(Q) = \underbrace{S\frac{D}{Q}}_{\text{ordering cost}} + \underbrace{H\frac{Q}{2}}_{\text{holding cost on avg inventory}}$$

But in the backorder EOQ, when the shipment of size Q arrives, it doesn't all become shelf stock, the first b units go to pay down whatever was backordered. So the peak on-hand level right after delivery is not Q anymore:

$$I_{\max} = Q - b$$

The cycle splits into two triangles instead of one:

$$t_1 = \frac{I_m}{D} \quad (\text{draining } I_m \to 0)$$
$$t_2 = \frac{b}{D} \quad (\text{accumulating backorders, } 0 \to b)$$

Full cycle:

$$t_1 + t_2 = \frac{I_{\max} + b}{D} = \frac{Q}{D} = T$$

Calculate two average-inventory terms area of the relevant triangle, divided by the whole cycle T:

- Positive Inventory

$$\overline{I} = \frac{\frac{I_m^2}{2D}}{\frac{Q}{D}} = \frac{I_m^2}{2Q}$$

- Negative Inventory

$$\overline{b} = \frac{\frac{b^2}{2D}}{\frac{Q}{D}} = \frac{b^2}{2Q}$$

Assemble the cost function
- Ordering cost: untouched
- Holding cost H now applies only to the on-hand average $\overline{I}$
- Penalty cost P (new) applies to the average backorder level $\overline{b}$

$$\begin{aligned}\mathrm{TVC}(Q,b) &= S\frac{D}{Q} + H\overline{I} + P\overline{b} \\ &= S\frac{D}{Q} + H\frac{(Q-b)^2}{2Q} + P\frac{b^2}{2Q}\end{aligned}$$

Check: set $b = 0$ (no backorders allowed). Then $I_{\max} = Q$ and the formula collapses to:

$$\begin{aligned}\mathrm{TVC}(Q,0) &= S\frac{D}{Q} + H\frac{Q^2}{2Q} + 0 \\ &= S\frac{D}{Q} + H\frac{Q}{2}\end{aligned}$$

Take gradient (partial derivatives) and set to 0:

$$\frac{\partial \mathrm{TVC}}{\partial b} = \frac{-H(Q-b) + Pb}{Q} = 0$$

$$\frac{\partial \mathrm{TVC}}{\partial Q} = -\frac{SD}{Q^2} + \frac{H}{2} - \frac{(H+P)b^2}{2Q^2} = 0$$

Solve $\frac{\partial \mathrm{TVC}}{\partial b} = 0$ for b:

$$b^* = \left(H\frac{Q}{H+P}\right)$$

Plug b into $\frac{\partial \mathrm{TVC}}{\partial Q}$ to solve for Q^* alone:

$$Q^* = \sqrt{\frac{2SD}{H}} \cdot \sqrt{\frac{H+P}{P}}$$

As $P \to \infty$, we get the normal EOQ (disallow backorders) (P is infinitely expensive)

Plug this Q^* into b^*:

$$b^* = \sqrt{\frac{2SDH}{P(H+P)}}$$

(a) *What is the EOQ for ACT in this case? Please show the steps you followed in deriving this quantity. (7 points)*

Hint: Set B the quantity backordered. Make a graph of how the policy will look like if backorders are allowed. Compute the time lengths T1 (demand is fully met) and T2 (demand is backordered) as functions of D, Q, and B. Compute average inventory and average backorders and construct the cost function (which is now a function of both Q and B). You can assume that the cost function is jointly convex to Q and B (i.e., you don't need to show second order derivative calculations). Then, take the partial derivatives of the cost function over Q and B. This will result to a system of two equations with two unknowns. By solving this system, you can obtain the optimal Q and B.

- Optimal order quantity: $Q^* =$ 715.94
- Optimal backorder level: $b^* =$ 257
- Peak on-hand inventory: $I_m^* = Q^* - b^* =$ 458.94
- Minimum annual variable cost: $\mathrm{TVC}(Q^*, b^*) =$ $\$1927.53$

(b) *Do you expect the new EOQ to be different (smaller or larger) from the classical EOQ derived in class? Please explain why. (3 points)*

Policy	Q	Annual variable cost
Classical EOQ (no backorders)	573.21	$2407.49
EOQ with backorders	715.94	$1927.53

